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Predicting Ten-year Coronary Heart Disease Risk with Machine Learning

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ABSTRACT

This report presents an end-to-end machine learning project that predicts whether a person will develop coronary heart disease (CHD) within ten years, using 4,238 records from the Framingham Heart Study and 15 clinical and behavioural variables. The workflow has seven steps: missing-value handling, visualisation, distribution analysis, feature engineering, dimensionality reduction, train/test splitting, and modelling. Six algorithms were compared, together with Stacking and soft-voting ensembles, tuned by randomised search and evaluated with repeated cross-validation. All models converged at a cross-validated ROC-AUC of about 0.73, and no algorithm outperformed another by more than statistical noise. Logistic regression was adopted as the final model because it is simple, well calibrated and interpretable; on the held-out test set it reached ROC-AUC = 0.699 (95% CI: 0.658–0.743). The results indicate that the performance ceiling is set by the information in the available variables rather than by the choice of algorithm, and that the model is not suitable for clinical decisions.

1. INTRODUCTION

Coronary heart disease is one of the leading causes of death worldwide, and early risk estimation enables preventive intervention. The Framingham Heart Study, begun in 1948, helped identify the classic risk factors such as age, blood pressure, smoking and diabetes (Dawber et al., 1951).

Problem statement: binary classification of whether a person will develop CHD within ten years (target variable TenYearCHD).

Objectives:

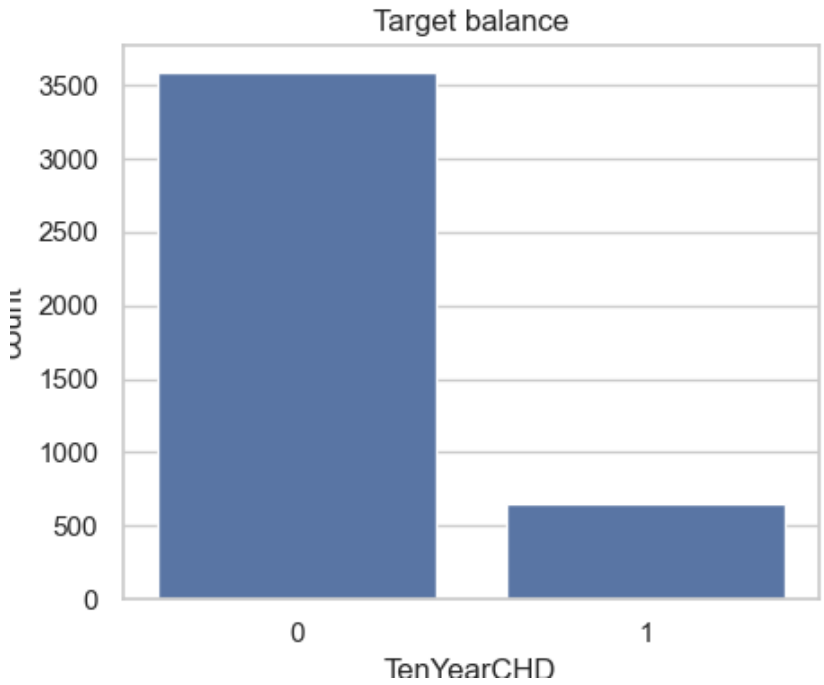
- Build a clean preprocessing pipeline that leaks no information from the test data.
- Compare several algorithms fairly, using cross-validation.
- Estimate performance honestly, with a confidence interval and metrics suited to class imbalance.
- Interpret the final model and state its limits.

Challenges: class imbalance (only 15.2% positive), weak signal between the predictors and the target, missing values, and outliers.

2. DATASET DESCRIPTION

The dataset contains 4,238 records and 16 columns: 15 predictors and the target. 15.2% of participants (about 644) developed CHD, versus 84.8% who did not. The predictors fall into four groups:

- **Demographic:** male (sex), age, education (level 1 to 4).
- **Behavioural:** currentSmoker, cigsPerDay (cigarettes per day).
- **Medical history and treatment:** BPMeds (blood-pressure medication), prevalentStroke, prevalentHyp (hypertension), diabetes.
- **Clinical measurements:** totChol (total cholesterol), sysBP and diaBP (systolic and diastolic blood pressure), BMI, heartRate, glucose.



3. GENERAL METHODOLOGY

- **Leakage prevention:** every statistic (median, clipping bounds, transform parameters, mean and standard deviation) is computed on the training data only, inside a Pipeline, and then applied to the test data.

- **Decisions by measurement:** every preprocessing option is evaluated by repeated 5-fold cross-validation and is never assumed to help. Differences smaller than the standard deviation (about 0.02) are treated as noise.

- **Evaluation metrics:** ROC-AUC is the main metric, complemented by PR-AUC, recall and precision. Accuracy is misleading here: a model that always predicts “no CHD” scores 84.8% without detecting a single case.

- **Environment:** Python 3.12 with pandas, scikit-learn, XGBoost, matplotlib and seaborn.

4. IMPLEMENTATION STEPS

4.1 Missing values:

We analysed the missingness and found 645 empty cells in seven columns, spread over 582 rows (13.7% of the data): glucose (388 values, 9.16%), then education (105), BPMeds (53), totChol (50), cigsPerDay (29), BMI (19) and heartRate (1). Missingness was unrelated to the outcome (CHD rate 14.9% in rows with missing values versus 15.2% in complete rows). Inspecting the data revealed logical rules, so a dedicated imputer (FraminghamImputer) was built:

- **BPMeds:** zero for people without hypertension (all 2,891 known cases among them are zero), and the most frequent value for the rest.
- **cigsPerDay:** all 29 missing values belong to smokers, so they were filled with the median of smokers (20 cigarettes), because the overall median is close to zero.
- **glucose:** the median of each diabetes group (about 78 for non-diabetics and 145 for diabetics).
- **education:** the most frequent value. totChol, BMI and heartRate: the median.

We also compared four general methods by cross-validation: the median, the median with a missing-value indicator column, KNN, and IterativeImputer.

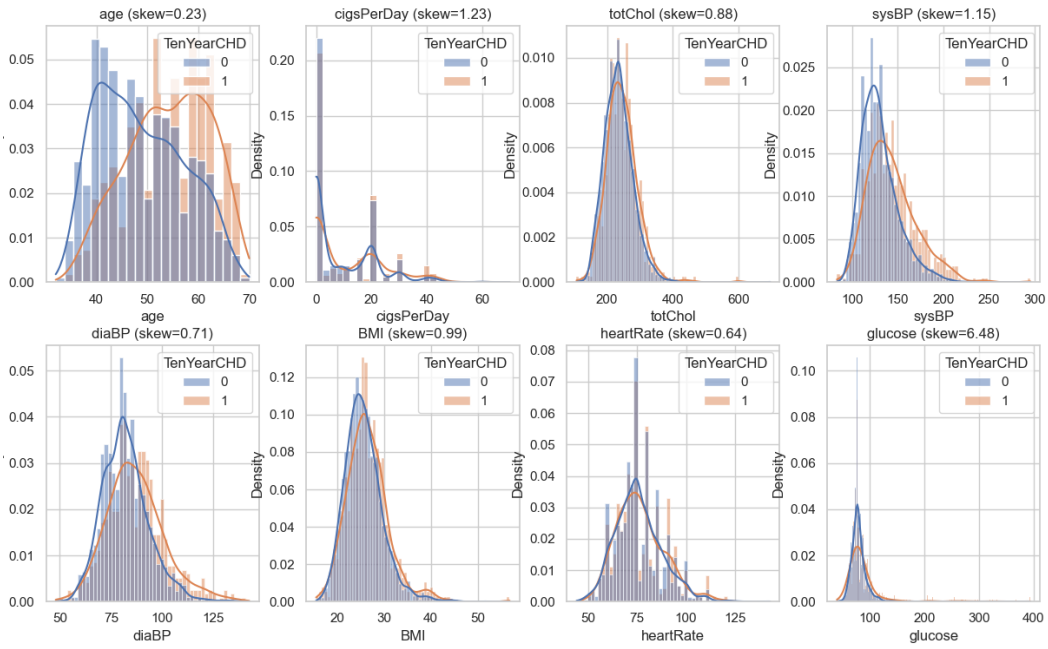
What we obtained. Missing cells dropped from 645 to zero without deleting any row. The four methods were practically identical (ROC-AUC between 0.7245 and 0.7248 for logistic regression), and the differences were far smaller than the standard deviation, so the rule-based imputer was adopted as the simplest and easiest to justify. Deleting incomplete rows was rejected because it would discard 13.7% of the data.

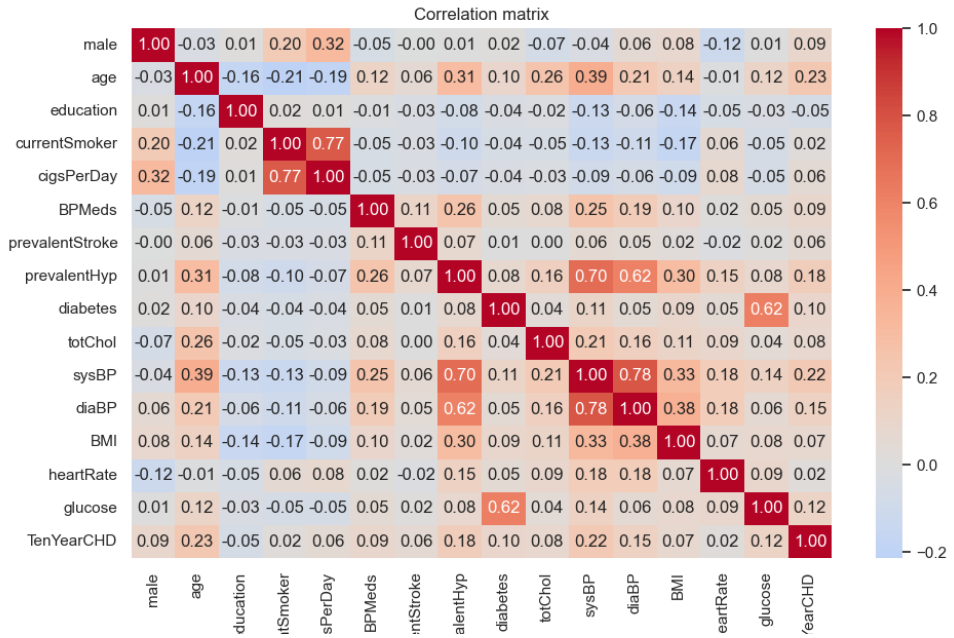
4.2 Data visualisation

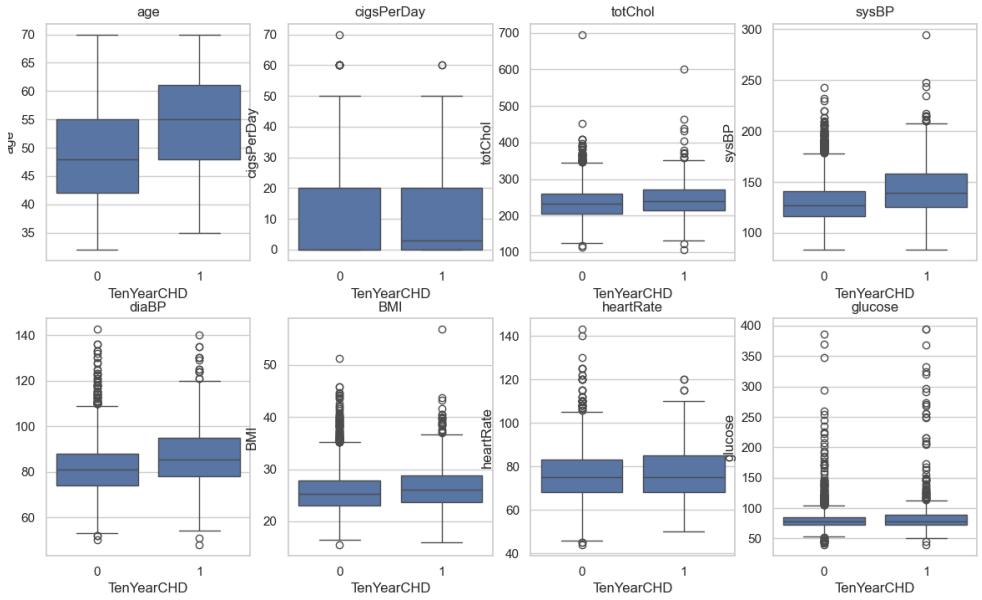
We plotted the distribution of every numeric variable by class (histograms), box plots, the CHD rate for each categorical variable, and the correlation matrix.

What we obtained.

- **Weak signal:** the strongest correlations with the target are age (0.225), sysBP (0.216), prevalentHyp (0.178), diaBP (0.145) and glucose (0.124); none exceeds 0.23.
- **Differences in CHD rate:** the rate is higher among people with diabetes (about 36% versus 14%), a previous stroke (about 43%), hypertension (about 24% versus 11%), and men (about 19% versus 12%), bearing in mind that some groups are small (about 25 stroke cases). These figures are approximate, read from the plots.
- **Correlated predictors:** sysBP and diaBP (0.78), currentSmoker and cigsPerDay (0.77), sysBP and prevalentHyp (0.70), diabetes and glucose (0.62).
- **Outliers (IQR rule):** glucose (6.3%), sysBP (3.0%) and BMI (2.3%); most are medically genuine (high glucose in diabetic patients), so they were not removed.







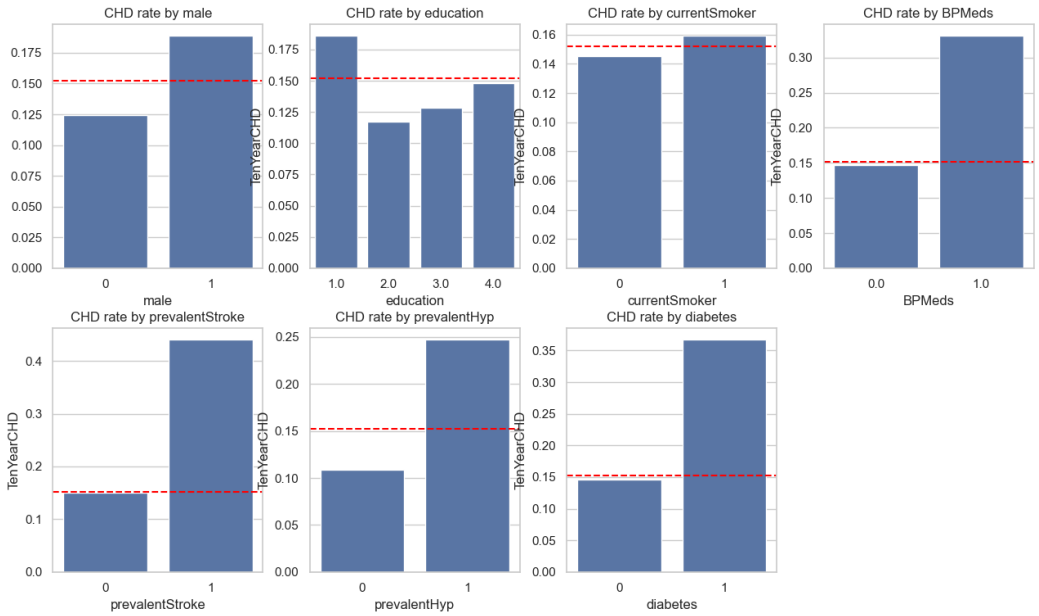
4.3 Data distribution

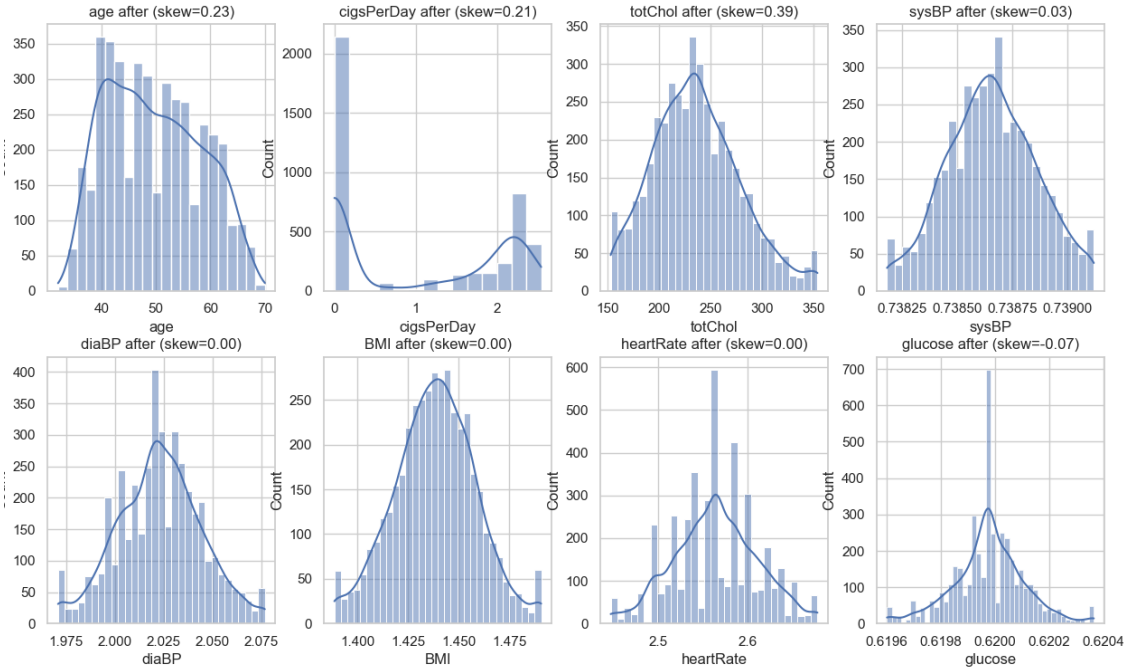
A transformer (DistributionTransformer) clips extreme values at the 1st and 99th percentiles in seven columns (winsorizing), then applies the Yeo-Johnson transformation (Yeo and Johnson, 2000) to the columns whose skewness remained above 0.5. Clipping affected 21 values in cigsPerDay, 56 in heartRate, and between 80 and 86 in each of totChol, sysBP, diaBP, BMI and glucose.

What we obtained. Skewness dropped in every treated column:

Column	Skewness before	Skewness after
glucose	6.48	-0.07
cigsPerDay	1.23	0.21
sysBP	1.15	0.03
BMI	0.99	0.00
totChol (clipping only)	0.88	0.39
diaBP	0.71	0.00
heartRate	0.64	0.00

cigsPerDay stayed bimodal (a large spike at zero for non-smokers and a second mode for smokers), because transformations cannot fix a distribution made of two groups; this is acceptable since currentSmoker already separates the two groups. After the transformation, values piled up at the range edges (the clipped values) and a spike appeared in the middle of glucose (the imputed values). The transformed values are small because they were not standardised; standardisation is applied later with StandardScaler.





4.4 Feature engineering

We created four groups of new features and tested each one alone and in combinations, by cross-validation (5 folds × 3 repeats) with logistic regression and HistGradientBoosting (HGB):

- **bp:** pulse pressure (sysBP − diaBP) and mean arterial pressure.
- **flags:** clinical indicators: blood pressure ≥ 140/90, glucose ≥ 126, BMI ≥ 30, cholesterol ≥ 240.
- **interactions:** age×sysBP, age×male, age×glucose, and a pack-years index (smoking × age).
- **risk_count:** the number of risk factors each person has.

Model	Baseline	Best group	Difference
Logistic regression	0.7238	flags: 0.7242	+0.0004
HGB	0.7007	interactions: 0.7027	+0.0020

The standard deviation across folds is 0.020–0.026, so the differences are noise, and no new features were adopted. The likely explanation is that the models already extract these relationships from the original columns, and that the clinical flags duplicate the information in prevalentHyp and sysBP.

4.5 Dimensionality reduction and feature selection

We checked multicollinearity with the variance inflation factor (VIF) and tried PCA (at 90%, 95% and 99% explained variance), SelectKBest (k=8 and k=11), selection by Random Forest importance, Lasso (Tibshirani, 1996), and dropping diaBP. Each was compared with no reduction by cross-validation, with the selection performed inside each fold to avoid leakage.

- **VIF:** only cigsPerDay (13.4) and currentSmoker (12.9) exceed the threshold of 5; sysBP (3.6), diaBP (2.9) and the remaining variables are below it.
- **PCA:** 11 of the 15 components are needed to reach 90% of the variance (92.1% in fact) and 12 to reach 95%, so little compression is possible. It also hurt performance: at 90%, HGB scored 0.6937 versus 0.7007, and logistic regression 0.7181 versus 0.7238.
- **SelectKBest and RF-importance selection:** also hurt (down to −0.0099 for logistic regression).
- **Lasso:** the only method that helped, and asymmetrically: HGB rose from 0.7007 to 0.7116 (+0.0109) while logistic regression was unchanged (−0.0001), a mean of +0.0054. It kept 9 features on the full data (male, age, cigsPerDay, BPMeds, prevalentStroke, prevalentHyp, diabetes, sysBP, glucose) and 10 on the training data alone (adding totChol), so the selection is unstable.
- It was re-tested in Step 7 across six models and gave a mean gain of +0.0026, narrowly above our adoption threshold of 0.002.

4.6 Train/test split

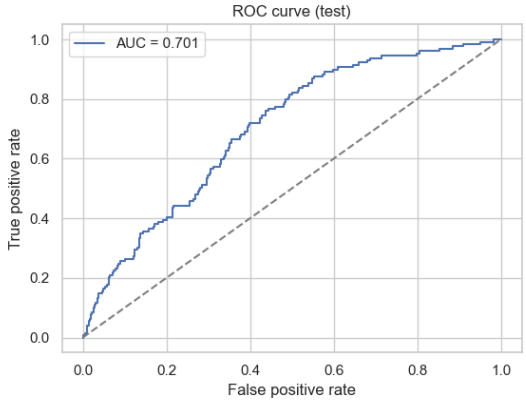
The data were split with stratification at 80/20 and random_state=42: 3,390 training records (515 positive) and 848 test records (129 positive), with almost identical positive rates (15.19% and 15.21%). We checked for leakage by comparing statistics computed on the training data alone with those computed on all data (for example, the median glucose of diabetic patients was 149 versus 145). Cross-validation used 15 folds (5×3) on the training data only. The baseline was computed: a model that always predicts the majority class scores an accuracy of 0.8479. The test set was locked until the final evaluation.

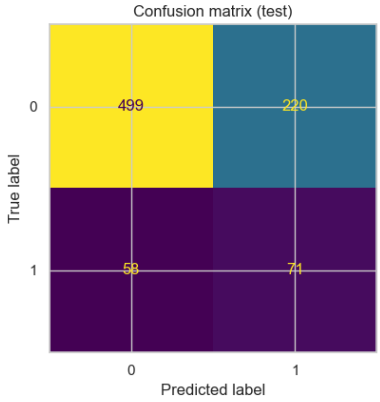
4.7 Modelling

We compared six algorithms: logistic regression, Random Forest, Extra Trees, HistGradientBoosting, SVM and XGBoost, using class weights to handle the imbalance. The hyperparameters of each model were tuned with RandomizedSearchCV (up to 20 trials per model, AUC with 5×2 cross-validation) (Bergstra and Bengio, 2012). The tuned models were then combined in two ways: Stacking (with a logistic-regression meta-learner) and soft voting. All candidates were compared with cross-validation on fresh folds, a model was selected by the results, and the decision threshold was set from out-of-fold predictions.

What we obtained (cross-validated ROC-AUC on the training data).

Model	Before tuning	After tuning	Final comparison
SVM	0.7044	0.7332	0.7314
Logistic regression	0.7291	0.7302	0.7305
Stacking	—	—	0.7297
Soft voting	—	—	0.7269
Extra Trees	0.7190	0.7266	0.7234
XGBoost	0.7095	0.7267	0.7220
HGB	0.7086	0.7235	0.7191
Random Forest	0.7200	0.7229	0.7175





The standard deviation is about 0.024. The spread among the top four models is only 0.0045, so they are statistically equivalent, and neither Stacking nor voting beat the single models. The automatic selection picked SVM, ahead of logistic regression by 0.0009, and it scored ROC-AUC = 0.7014 on the test set (95% CI 0.661–0.746).

Final decision: Logistic regression was adopted because it is statistically equivalent to SVM, gives interpretable probabilities and medically readable coefficients, and is the simplest and least prone to overfitting. Class weights were removed from the final model so that it outputs realistic probabilities (cross-validated AUC 0.7293 versus 0.7305 with weights), and the threshold was set at 0.163 using Youden's criterion (Youden, 1950) on out-of-fold predictions (recall = 0.664 and precision = 0.282 at that threshold on the training data).

5. FINAL RESULTS OF THE ADOPTED MODEL

The logistic regression was evaluated on the 848 test records:

Metric	Value
ROC-AUC	0.699 (95% CI: 0.658–0.743)
PR-AUC	0.305 (random model: 0.152)
Brier score	0.121 (always predicting the overall rate: 0.129)
Sensitivity (recall) for CHD	0.53 (68 of 129)
Precision	0.24
Specificity	0.70 (501 of 719)
Accuracy	0.67 (baseline 0.848)

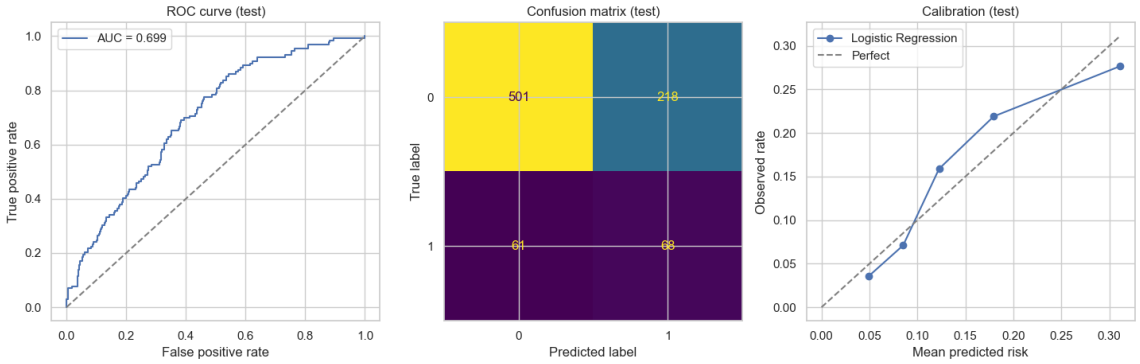
Confusion matrix: 501 true negatives, 218 false positives, 61 false negatives and 68 true positives. The model flags about 34% of people as high risk, and roughly three of every four alerts are false alarms.

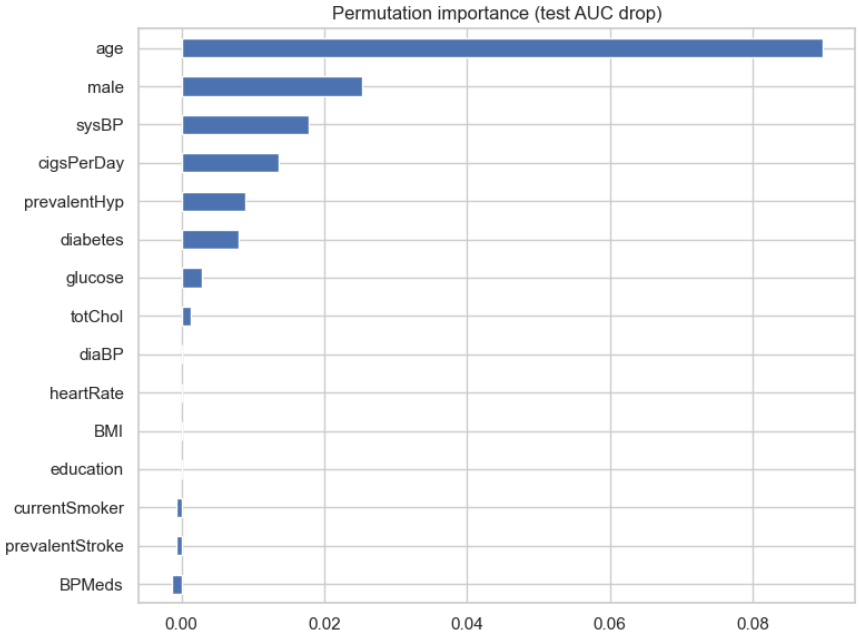
Calibration: The predicted probabilities track the observed rates across five risk groups:

Risk group	Predicted	Observed
Lowest	0.049	0.035
Low	0.085	0.071
Middle	0.123	0.159
High	0.179	0.219
Highest	0.311	0.276

Coefficients (odds ratio per standard deviation):

Variable	Coefficient	Odds ratio
age	0.460	1.58
sysBP	0.249	1.28
cigsPerDay	0.201	1.22
male	0.199	1.22
prevalentHyp	0.128	1.14
diabetes	0.091	1.10
totChol	0.084	1.09
glucose	0.082	1.09
prevalentStroke	0.074	1.08
BPMeds	0.068	1.07





6. DISCUSSION

- **Performance ceiling:** All algorithms, despite their different structures, converged at an AUC between 0.72 and 0.73. This suggests that the barrier lies in the information in the available variables, not in the algorithm or its tuning, which is why a wider hyperparameter search and ensembling did not help.
- **Test-set gap:** The test AUC (0.70) is lower than the cross-validated one (about 0.73). The difference lies within the confidence interval; part of it reflects optimism from selection and tuning on the training data, and part the small test set (129 positives). The more honest estimate of expected performance is about 0.70.
- **Consistent interpretation:** Age is by far the strongest factor, followed by systolic blood pressure, smoking, sex, hypertension and diabetes, in line with the known risk factors in the Framingham literature (Wilson et al., 1998). These are associations, not causal effects.
- **Overall accuracy:** The accuracy of 0.67 is below the 0.848 baseline, which is not a flaw: the threshold was chosen to balance sensitivity and specificity, and raising accuracy to 0.85 would mean missing most of the patients.
- **Calibration:** Because the probabilities are realistic, the model is better suited to ranking people by risk than to making a binary decision.

7. LIMITATIONS

1. **The test set was seen twice:** SVM was evaluated on it first, then logistic regression. The decision to adopt logistic regression was based on cross-validation results and interpretability, but the test set is no longer fully hidden, so its figures are closer to a conditional estimate than a fully independent one.
2. **Tuning:** the regularisation parameter C of logistic regression was tuned with class weights, which were then removed without re-tuning C (the expected effect is small).
3. **Lasso feature selection is marginal and unstable** (mean gain +0.0026).
4. **Single imputation** of missing values does not reflect the uncertainty in them; multiple imputation would be more accurate.
5. **Test-set size:** only 129 positives, which widens the confidence interval.
6. **No external validation,** and medically important variables are absent (HDL and LDL cholesterol, family history, other medications).
7. **The relationships are associations,** not causal, and the model is unsuitable for clinical decisions (alert precision of 24%).

8. CONCLUSION AND RECOMMENDATIONS

A complete seven-step pipeline was built, applying one rule to every decision: nothing is adopted before its effect is shown by cross-validation. Most of the advanced processing (feature engineering, dimensionality reduction, model combination) added nothing beyond noise, and the simple logistic regression (AUC = 0.699 on the test set) proved sufficient, calibrated and interpretable.

Recommendations:

- Add new, medically relevant variables, the most likely route to a real improvement.
- Validate externally on an independent dataset.
- Use multiple imputation, and re-tune C without class weights.
- Explore survival-analysis models (such as Cox regression), since the target is time-dependent.
- Keep a test set that is used only once, for a single model.